

5

A

Consider the whole system,

$$F - 6f = 6Ma$$

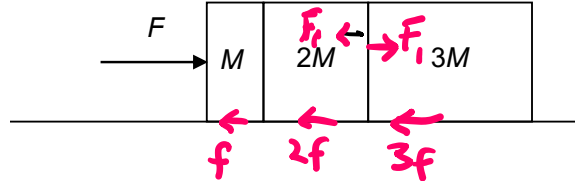
$$a = \frac{F - 6f}{6M}$$

Let the force that $3M$ acts on $2M$ be F_1 .

Consider Newton's 2nd law on $3M$,

$$F_1 - 3f = 3Ma$$

$$F_1 = 3M \left(\frac{F - 6f}{6M} \right) + 3f = \frac{F}{2}$$



6

A

Apply COM to the collision: